



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

- State the Stokes' Theorem.
- Verify that $w_{xy} = w_{yx}$ if $w = \ln(2x + 3y)$.
- Find unit tangent $\underline{T}$ if $\underline{r}(t) = (\cosh t)\hat{i} - (\sinh t)\hat{j} + t\hat{k}$.
- Convert the Cartesian equation $xy = 2$ in polar form.
- Find the center and radius of the sphere $x^2 + y^2 + z^2 + x - 4z + 1 = 0$.
- The line through the point $P(3, -4, -1)$ parallel to the vector $\hat{i} + \hat{j} + \hat{k}$.

Answer the following questions

Q. No. 2 Show that if f is homogeneous of degree n , show that

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f(x, y). \quad [7]$$

Q. No. 3 If $y = e^{c_1 x_1 + c_2 x_2 + c_3 x_3 + \dots + c_n x_n}$, where $c_1^2 + c_2^2 + c_3^2 + \dots + c_n^2 = 1$, show that

$$\frac{\partial^2 y}{\partial x_1^2} + \frac{\partial^2 y}{\partial x_2^2} + \frac{\partial^2 y}{\partial x_3^2} + \dots + \frac{\partial^2 y}{\partial x_n^2} = y. \quad [7]$$

Q. No. 4 Verify that the function $u = 1/\sqrt{x^2 + y^2 + z^2}$ is a solution of the three dimensional Laplace equation $u_{xx} + u_{yy} + u_{zz} = 0$. [8]

Q. No. 5 Convert the Cartesian equation $xy = 2$ in polar form. [8]

Solved PP

Short Answers

① Stroke's Theorem ✓

→ Stokes theorem is fundamental theorem in vector calculus.

→ It established a relationship b/w surface integral & line integrals, which is generalization of the fundamental theorem of calculus to higher dimensions.

Mathematically Stokes' theorem:

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

$\mathbf{F}$ → is a vector field

∂S → boundary curve of surface S .

$d\mathbf{r}$ → is a vector representing an infinitesimal element of the boundary curve.

$\nabla \times \mathbf{F}$ → represents the curl of vector field $\mathbf{F}$

$d\mathbf{S}$ → vector represent element of surface.

②

Verify that

$$w_{xy} = w_{yx} \text{ if } w = \ln(2x+3y)$$

$$w_{xy}$$

$$w_{yx}$$

take partial diff w.r.t 'x'

$$f(x,y) = \ln(2x+3y)$$

$$\frac{\partial f}{\partial x} = \frac{1}{2x+3y} \cdot (2x+3y)'$$

$$= \frac{1}{2x+3y} \cdot 2$$

$$w_{xy} = \frac{2}{2x+3y}$$

Partial diff w.r.t 'y'

$$\frac{\partial f}{\partial y} = \frac{1}{2x+3y} \cdot (2x+3y)'$$

$$= \frac{1}{2x+3y} \cdot 3$$

$$= \frac{3}{2x+3y} \Rightarrow \boxed{\frac{3}{2x+3y}}$$

these wry $\neq$ wry is not
equal to each other

③ Find Unit tangent

T if $r(t) = (\cosh t)\hat{i} - (\sinh t)\hat{j} + t\hat{k}$

→ Derivation $r(t)$

$$= \frac{d}{dt} [(\cosh t)\hat{i} - (\sinh t)\hat{j} + t\hat{k}]$$

$$r'(t) = (\sinh t)\hat{i} - (\cosh t)\hat{j} + \hat{k}$$

→ magnitude of $r(t)$

$$\|r(t)\| = \sqrt{(\sinh t)^2 + (-\cosh t)^2 + (1)^2}$$

$$\|r'(t)\| = \sqrt{(\sinh^2 t) + (\cosh^2 t) + 1}$$

→ using hyperbolic identity

$$\boxed{\cosh^2 t - \sinh^2 t = 1}$$

$$\|r(t)\| = \sqrt{(\cosh^2 t - 1) + (\cosh^2 t) + 1}$$

lay

$$|\gamma(t)| = \sqrt{2 \cosh^2 t}$$

$$|\gamma(t)| = \sqrt{2} \cosh t$$

→ Normalize the tangent vector $\gamma(t)$

$$T = \frac{\gamma'(t)}{|\gamma'(t)|}$$

$$T = \frac{(\sinh t)i - (\cosh t)j + k}{\sqrt{2} \cosh t}$$

$$= \left(\frac{\sinh t}{\sqrt{2} \cosh t} \right) i - \left(\frac{1}{\sqrt{2}} \right) j + \left(\frac{1}{\sqrt{2} \cosh t} \right) k$$

$$T = \left(\frac{\sinh t}{\sqrt{2} \cosh t} \right) i - \left(\frac{1}{\sqrt{2}} \right) j + \left(\frac{1}{\sqrt{2} \cosh t} \right) k$$

④

Convert
Cartesian equation to Polar
form

$$xy = 2$$

→ Polar form coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$xy = 2$$

$$(r \cos \theta)(r \sin \theta) = 2$$

$$r^2 \cos \theta \sin \theta = 2$$

use trigonometric identity

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

Now

$$\frac{1}{2} r^2 \sin 2\theta = 2$$

$$r^2 \sin 2\theta = 2 \times 2$$

$$\boxed{r^2 \sin 2\theta = 4} \quad \sqrt{}$$

⑤ Find center & Radius of sphere

$$x^2 + y^2 + z^2 + x - 4z + 1 = 0$$

Sphere equation

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

→

$$\text{For } x \rightarrow x^2 + x$$

$$= x^2 + x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$

$$= \left(x + \frac{1}{2}\right)^2 - \frac{1}{4}$$

$$\text{For } y: y^2$$

$$\text{For } z: z^2 - 4z$$

$$z^2 - 4z + 4 - 4 = (z-2)^2 - 4$$

Now substitute ↙

$$\left(x + \frac{1}{2}\right)^2 - \frac{1}{4} + y^2 + (z-2)^2 - 4 + 1 = 0$$

$$\left(x + \frac{1}{2}\right)^2 + y^2 + (z-2)^2 - \frac{1}{4} - 4 + 1 = 0$$

$$\left(x + \frac{1}{2}\right)^2 + y^2 + (z-2)^2 = \frac{5}{4}$$

→ identify the center & radius

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

center (h, k, l) is $\boxed{\left(-\frac{1}{2}, 0, 2\right)}$

Radius r is $\sqrt{\frac{5}{4}} \Rightarrow \boxed{\frac{\sqrt{5}}{2}}$

⑥ line through the point
 $P(3, -4, -1)$ parallel to
vector $\hat{i} + \hat{j} + \hat{k}$.

$$r(t) = r_0 + tv$$

$$r(t) = (3\hat{i} - 4\hat{j} + \hat{k}) + t(\hat{i} + \hat{j} + \hat{k})$$

$$r(t) = (3+t)\hat{i} + (-4+t)\hat{j} + (1+t)\hat{k}$$

$$x(t) = 3+t$$

$$y(t) = -4+t$$

$$z(t) = 1+t$$

①

Long Question

Show that f is homogenous of degree n , then

$$\left[x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} \right] = n(n-1)f(x,y)$$

Sol $\rightarrow$

Homogeneous degree n

$$f(\lambda x, \lambda y) = \lambda^n f(x, y)$$

$$\frac{\partial}{\partial \lambda} [f(\lambda x, \lambda y)] = \frac{\partial}{\partial \lambda} [\lambda^n f(x, y)]$$

$$= n \lambda^{n-1} f(x, y)$$

multiply by λ

$$\lambda x \frac{\partial f}{\partial (\lambda x)} + \lambda y \frac{\partial f}{\partial (\lambda y)} = n \lambda^n f(x, y)$$

substitute $\lambda = 1$

$$n^2 \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial y^2} = n f(x, y)$$

→ here take 1 → 4 time
diff then got

$$n^2 \frac{\partial^2 f}{\partial x^2} + 2ny \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = (n-1) n f(x, y)$$

(2)

$$y = e^{c_1 x_1 + c_2 x_2 + c_3 x_3 + \dots + c_n x_n}$$

$$\text{where } c_1^2 + c_2^2 + c_3^2 + \dots + c_n^2 = 1$$

Show that

$$\frac{\partial^2 y}{\partial x_1^2} + \frac{\partial^2 y}{\partial x_2^2} + \dots + \frac{\partial^2 y}{\partial x_n^2} = y$$

Solution

$$\frac{\partial y}{\partial x_1} = c_1 \cdot e^{c_1 x_1 + c_2 x_2 + \dots + c_n x_n} = c_1 y$$

$$\frac{\partial y}{\partial x_2} = c_2 \cdot e^{c_1 x_1 + c_2 x_2 + \dots + c_n x_n} = c_2 y$$

$$\frac{\partial y}{\partial x_n} = c_n \cdot e^{c_1 x_1 + c_2 x_2 + \dots + c_n x_n} = c_n y$$

$$\frac{\partial y}{\partial x_n} = c_n \cdot e^{c_1 x_1 + c_2 x_2 + \dots + c_n x_n}$$

Second order Partial Derivation

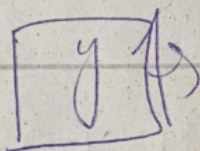
$$\frac{\partial^2 y}{\partial x_1^2} = c_1^2 + e^{c_1 x_1 + c_2 x_2 + \dots + c_n x_n} = c_1^2 y$$

$$\frac{\partial^2 y}{\partial x_2^2} = c_2^2 + e^{c_1 x_1 + c_2 x_2 + \dots + c_n x_n} = c_2^2 y$$

$$\frac{\partial^2 y}{\partial x_n^2} = c_n^2 + e^{c_1 x_1 + c_2 x_2 + \dots + c_n x_n} = c_n^2 y$$

sum up of Partial Derivation

$$\frac{\partial^2 y}{\partial x_1^2} = \frac{\partial^2 y}{\partial x_2^2} + \dots + \frac{\partial^2 y}{\partial x_n^2} = 1^* y = y$$



$$(x^2 + y^2 + z^2)^{-3/2}$$

$$(x^2 + z^2)^{-3/2}$$

⑨ Verify the function

$u = \frac{1}{\sqrt{x^2+y^2+z^2}}$ is a solution
of the three dimensional
laplace equation

$$u_{xx} + u_{yy} + u_{zz} = 0$$

Solution

$$\frac{\partial u}{\partial x} = - \frac{x}{(x^2+y^2+z^2)^{3/2}}$$

$$= - \frac{x}{(x^2+y^2+z^2)^{3/2}} \rightarrow \boxed{-x(x^2+y^2+z^2)^{-3/2}}$$

$$\frac{\partial u}{\partial y} = - \frac{y}{(x^2+y^2+z^2)^{3/2}}$$

$$\boxed{-y(x^2+y^2+z^2)^{-3/2}}$$

$$\frac{\partial u}{\partial z} = - \frac{z}{(x^2+y^2+z^2)^{3/2}} = -z(x^2+y^2+z^2)^{-3/2}$$

$$\boxed{-z(x^2+y^2+z^2)^{-3/2}}$$

2nd Partial Derivation

equation

$$\frac{\partial^2 u}{\partial x^2} = -(x^2 + y^2 + z^2)^{-3/2} + 3x^2(x^2 + y^2 + z^2)^{-5/2}$$

$$\frac{\partial^2 u}{\partial y^2} = -(x^2 + y^2 + z^2)^{-3/2} + 3y^2(x^2 + y^2 + z^2)^{-5/2}$$

$$\frac{\partial^2 u}{\partial z^2} = -(x^2 + y^2 + z^2)^{-3/2} + 3z^2(x^2 + y^2 + z^2)^{-5/2}$$

identity

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -3(x^2 + y^2 + z^2)^{-3/2} + 3(x^2 + y^2 + z^2)(x^2 + y^2 + z^2)^{-5/2}$$

$$= \boxed{-3(x^2 + y^2 + z^2)^{-3/2} + 3(x^2 + y^2 + z^2)^{-3/2} = 0} \quad \boxed{1/2}$$

④

Convert Cartesian equation
to Polar Form

$$\boxed{xy = 2}$$

lets

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$xy = 2$$

$$(r \cos \theta)(r \sin \theta) = 2$$

$$r^2 \cos \theta \sin \theta$$

use trigonometric identity

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

Now

$$\frac{1}{2} r^2 \sin 2\theta = 2$$

2

$$r^2 \sin 2\theta = 2 \times 2$$

$$\boxed{r^2 \sin 2\theta = 4} \quad \uparrow$$

Keep Shine